
Data Preprocessing

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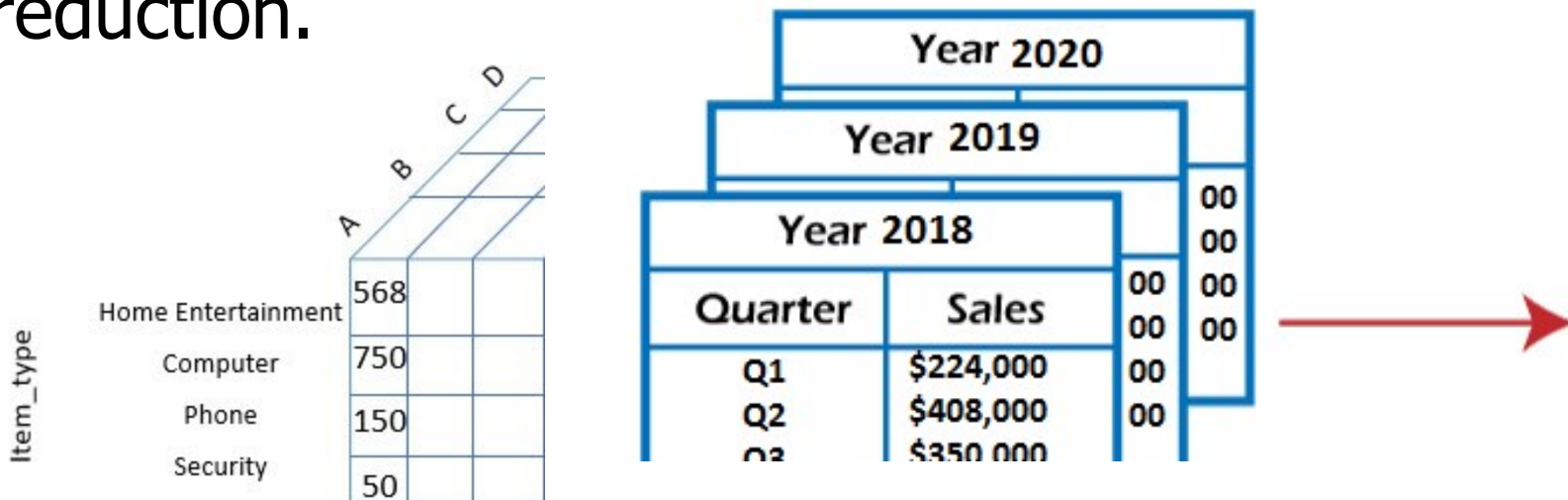
- Why preprocess the data?
- Data cleaning
- Data integration and transformation
- Data reduction
- Discretization and concept hierarchy generation
- Summary

Data Reduction Strategies

- Why data reduction?
 - A database/data warehouse may store terabytes of data
 - Complex data analysis/mining may take a very long time to run on the complete data set
- Data reduction
 - Obtain a reduced representation of the data set that is much smaller in volume but yet produce the same (or almost the same) analytical results
- Data reduction strategies
 - Data cube aggregation
 - Dimensionality reduction
 - Data Compression
 - Numerosity reduction — e.g., fit data into models
 - Discretization and concept hierarchy generation

Data Cube Aggregation

- This method is used to condense data into a more manageable format.
- It is a multidimensional aggregation that represents the original data set by aggregating at multiple layers of a data cube, resulting in data reduction.



Dimensionality reduction

- Dimensionality reduction is the process of reducing the number of features in a dataset while retaining as much information as possible.
- This can be done to reduce the complexity of a model, improve the performance of a learning algorithm, or make it easier to visualize the data.
- Principal Component Analysis (PCA), Factor Analysis (FA), Linear Discriminant Analysis (LDA) and Truncated Singular Value Decomposition (SVD) are some examples of linear dimensionality reduction methods.

Feature Selection

- Feature Selection method selects a subset of relevant features while keeping the original feature space intact.
- The focus is on identifying the most informative features for modeling.
- While dimensionality Reduction projects the data onto a lower-dimensional space by transforming the feature space.

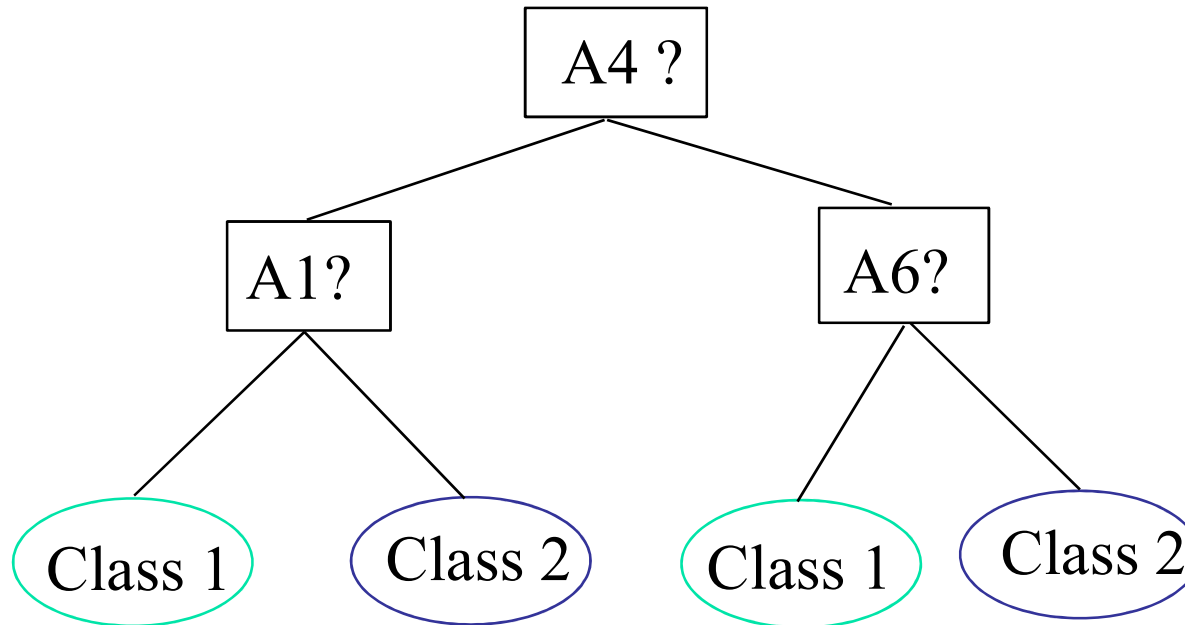
Feature selection

- Feature selection (i.e., attribute subset selection):
 - Select a minimum set of features such that the probability distribution of different classes given the values for those features is as close as possible to the original distribution given the values of all features.
- Heuristic methods (due to exponential # of choices):
 - Step-wise forward selection
 - Step-wise backward elimination
 - Combining forward selection and backward elimination
 - Decision-tree induction

Example of Decision Tree Induction

Initial attribute set:

{A1, A2, A3, A4, A5, A6}



-----> Reduced attribute set: {A1, A4, A6}

Heuristic Feature Selection Methods

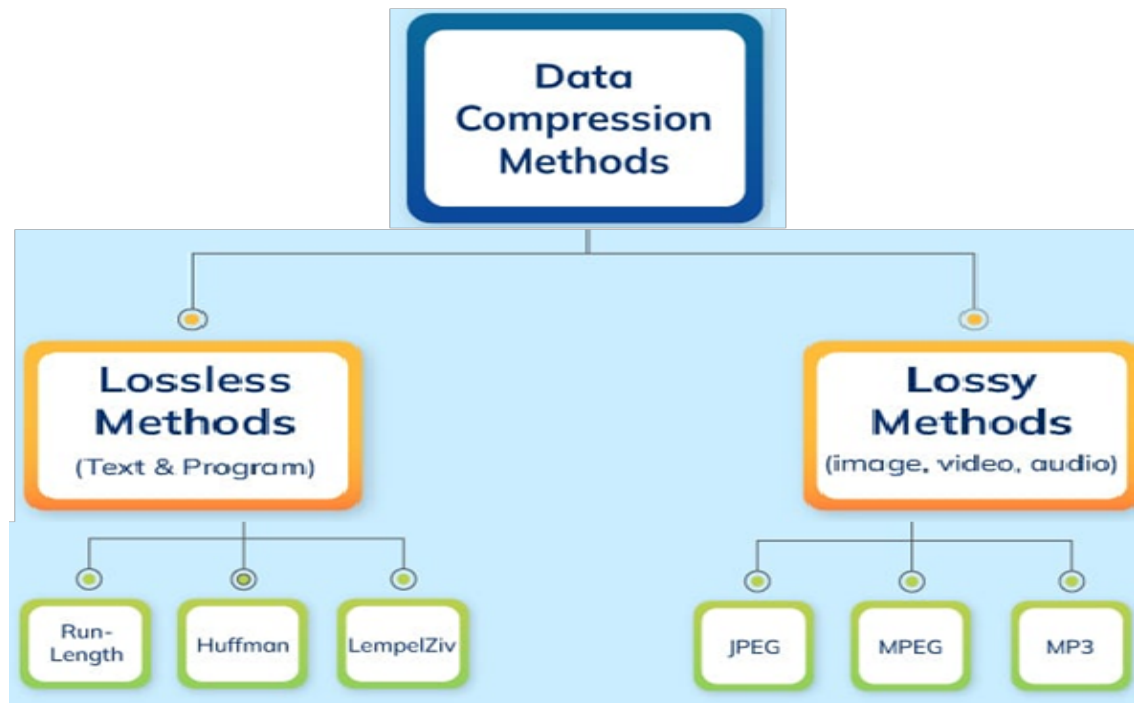
- There are 2^d possible sub-features of d features
- Several heuristic feature selection methods:
 - Best single features under the feature independence assumption: choose by significance tests
 - Best step-wise feature selection:
 - The best single-feature is picked first
 - Then next best feature condition to the first, ...
 - Step-wise feature elimination:
 - Repeatedly eliminate the worst feature
 - Best combined feature selection and elimination
 - Optimal branch and bound:
 - Use feature elimination and backtracking

Data compression

- Whether you've just started out with digital photography, compression is important, because digital images can be enormous in their original size – making them far harder to handle.
- Data compression is the process of modifying, encoding or converting the bits structure of data in such a way that it consumes less space on disk.
- It enables reducing the storage size of one or more data instances or elements.
- Data compression is also known as source coding or bit-rate reduction.

Data Compression

- There are two methods of compression –lossless and lossy. **Lossless** reduces file size by removing unnecessary metadata. **Lossy** reduces file size by permanently removing some of the original data.



Lossless Compression

- Lossless compression reduces file size by removing unnecessary metadata.
- With lossless compression the file data is restored and rebuilt in its original form after decompression, enabling the image to take up less space without any discernible loss in picture quality.
- Advantages: No loss in quality. Slight decreases in file sizes.
- Disadvantages: Compressed files are larger than lossy files.

Lossless Compression

- No data is lost and as the process can be reversed, it's also known as reversible compression.
- The image with GIF (Graphics Interchange File) format is generally compressed using lossless compression.
- RAW, BMP and PNG are also lossless image formats.
- **Some Lossless compression techniques are:**
 - (i) RLE (Run Length Encoding) (ii) Content Mixing
 - (iii) Huffman Encoding (iv) Adaptive Huffman Coding
 - (v) Shannon Fano Encoding.

Lossy Compression

- Lossy reduces file size by permanently removing some of the original data.
- Here, an algorithm scans image files and reduces their size by discarding information considered less important or undetectable to the human eye.
- Advantage - Small file sizes. Ideal for web use. Lots of tools, plugins and software support it.
- Disadvantage: Quality degrades due to higher rate of compression.

Lossy Compression

- Using lossy methods therefore requires you to make a balanced judgement between:
 - i) storage/delivery requirements ii) loading times (e.g. on the web) iii) image quality
- Some lossy compression techniques are: Discrete Cosine Transform (DCT), Discrete wavelet transform, fractal compression, transform encryption, etc.

Difference between Lossless and Lossy compression

1. What it does? **Lossy** – Permanently removes file data
Lossless - Restores and rebuilds compressed data
2. When it is used? **Lossy** – When file information loss is acceptable. **Lossless** - When file information loss is unacceptable
3. Applications: **Lossy** - Images, video, audio
Lossless - Text, images, audio
4. File types: **Lossy** - Images: JPEG, Video: MPEG, AVC, HEVC, Audio: MP3, AAC
Lossless - Images: RAW, BMP, PNG, General: ZIP, Audio: WAV, FLAC

Numerosity Reduction

- Parametric methods
 - Assume the data fits some model, estimate model parameters, store only the parameters, and discard the data (except possible outliers)
 - Example: Regression, Log-linear models
- Non-parametric methods
 - Do not assume models
 - Example: histograms, clustering, sampling, data cube aggregation.

Numerosity Reduction

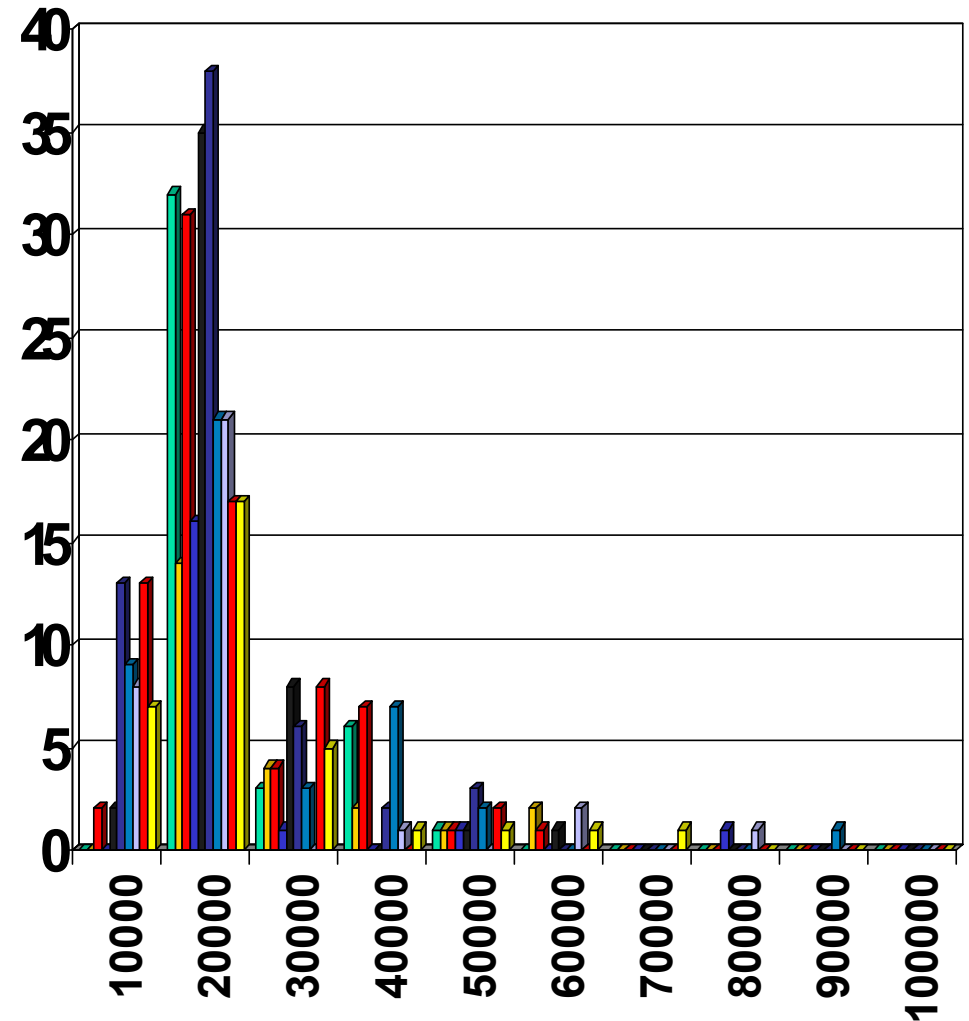
- Reduce data volume by choosing alternative, smaller forms of data representation. Different numerosity reduction techniques are:
- **Data Sampling:** This technique involves selecting a subset of the data points to work with, rather than using the entire dataset. This can be useful for reducing the size of a dataset while still preserving the overall trends and patterns in the data.
- **Clustering:** This technique involves grouping similar data points together and then representing each group by a single representative data point.
- **Data Aggregation:** This technique involves combining multiple data points into a single data point by applying a summarization function.
- **Data Generalization:** This technique involves replacing a data point with a more general data point that still preserves the important information.

Reduction using Regression and Log-Linear Models

- Linear regression: Data are modeled to fit a straight line
 - Often uses the least-square method to fit the line
- Multiple regression: allows a response variable Y to be modeled as a linear function of multidimensional feature vector
- Log-linear model: approximates discrete multidimensional probability distributions

Reduction using Histograms

- Divide data into buckets and store average (sum) for each bucket
- Partitioning rules:
 - Equal-width: equal bucket range
 - Equal-frequency (or equal-depth)



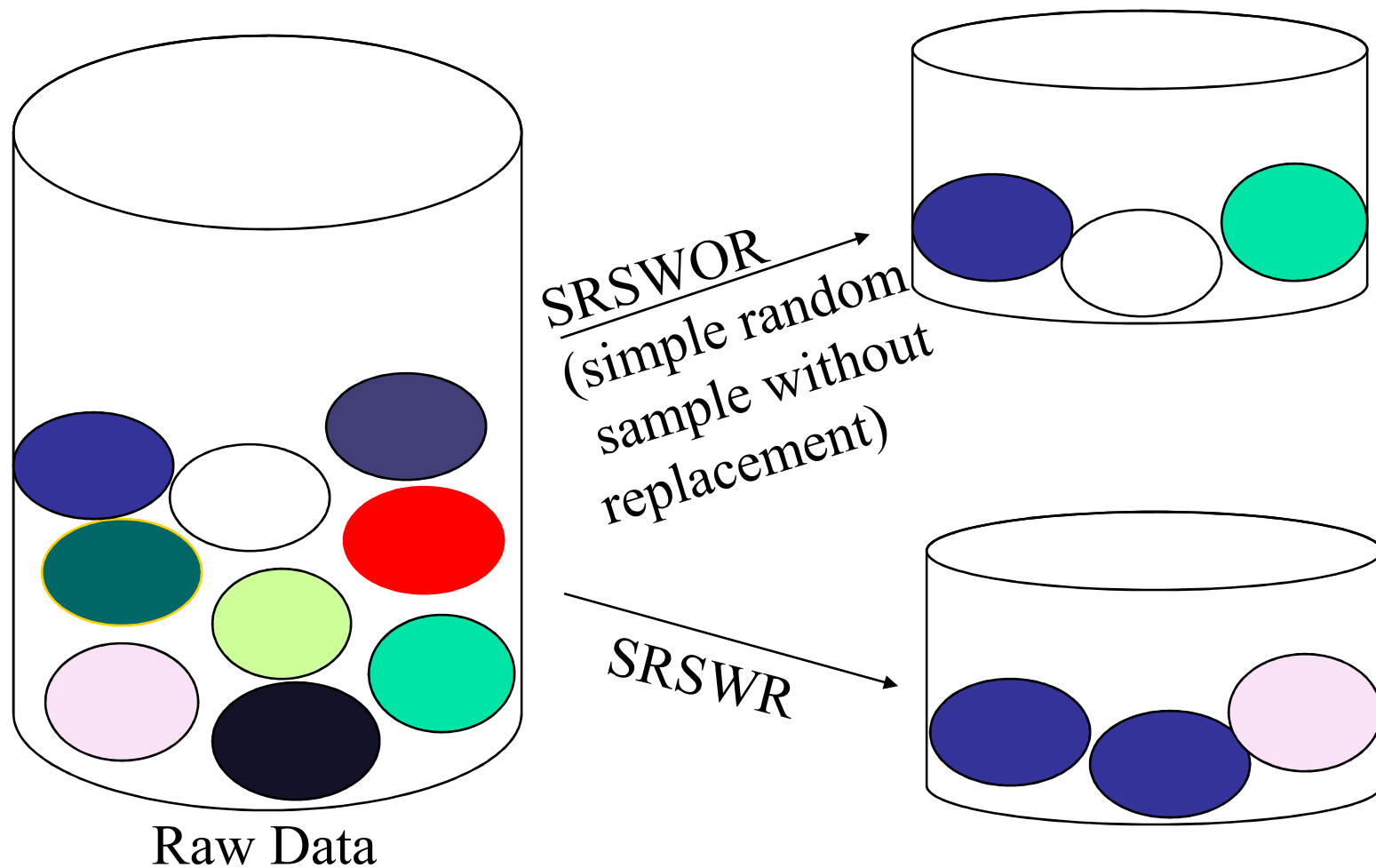
Reduction using Clustering

- Partition data set into clusters based on similarity, and store cluster representation (e.g., centroid and diameter) only
- Can be very effective if data is clustered but not if data is “smeared”
- Can have hierarchical clustering and be stored in multi-dimensional index tree structures
- There are many choices of clustering definitions and clustering algorithms
- Cluster analysis will be studied later in depth

Reduction using Sampling

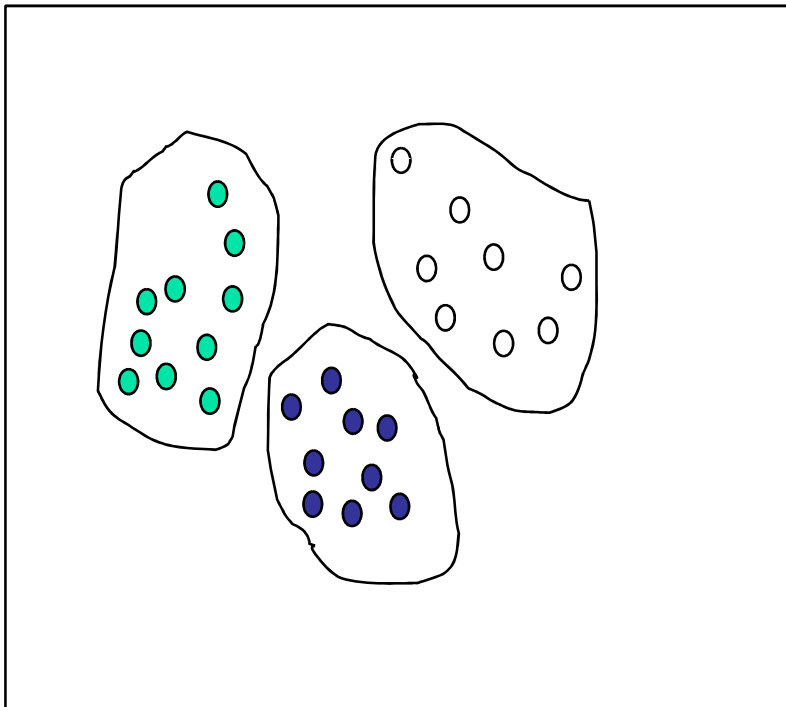
- Sampling: obtaining a small sample s to represent the whole data set N
- Allow a mining algorithm to run in complexity that is potentially sub-linear to the size of the data
- Choose a **representative** subset of the data
 - Simple random sampling may have very poor performance in the presence of skew
- Develop adaptive sampling methods
 - Stratified sampling:
 - Approximate the percentage of each class (or subpopulation of interest) in the overall database
 - Used in conjunction with skewed data

Sampling: with or without Replacement

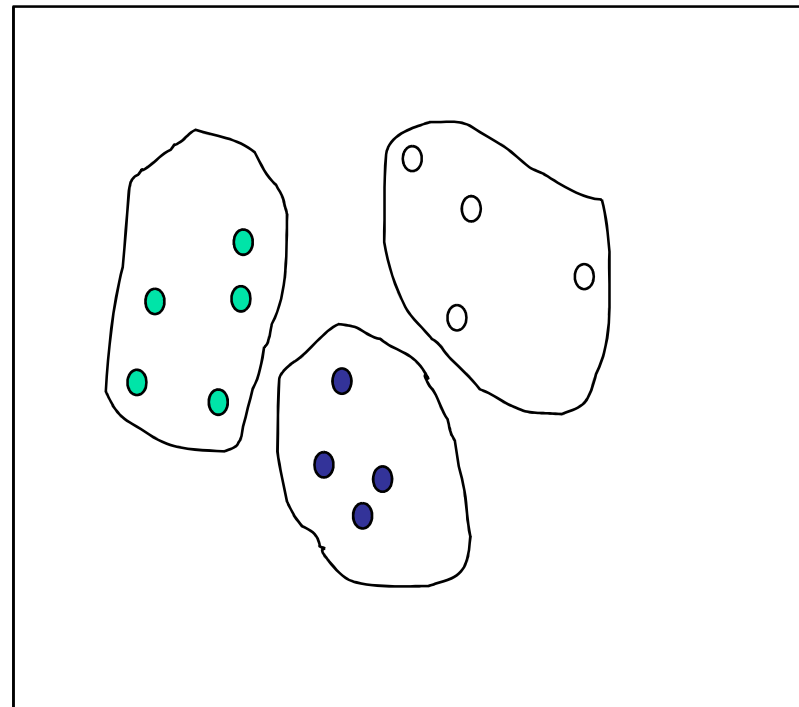


Sampling: Cluster or Stratified Sampling

Raw Data



Cluster/Stratified Sample



Numerosity Reduction

- In conclusion, numerosity reduction can have both advantages and disadvantages.
- It can improve the efficiency of machine learning algorithms by reducing the number of data points in a dataset. However, it can also result in a loss of information and make it harder to interpret the results.
- So the numerosity reduction can have a trade-off between the accuracy and the size of the data. The more data points are reduced, the less accurate the model will be and the less generalizable it will be.
- It's important to weigh the pros and cons of numerosity reduction and carefully assess the risks and benefits before implementing it.

Data Preprocessing

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- Data cleaning
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- Discretization and concept hierarchy generation
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Attributes and Measurement

- Here, we address the issue of describing data by considering what types of attributes are used to describe data objects.
- What is an attribute?
 - An attribute is a property of an object that may vary, either from one object to another or from one time to another. For ex, eye color varies from person to person, while the temperature of an object varies over time.

Measurement

- What is a measurement?
 - A measurement scale is a rule (function) that associates a numerical or symbolic value with an attribute of an object.
 - The process of measurement is the application of a measurement scale to associate a value with a particular attribute of a specific object.
 - For instance, we classify someone as male or female, we count the number of chairs in a room. In all cases, the “physical value” of an attribute of an object is mapped to a numerical or symbolic value.

Type of an Attribute

- Generally, we define four types of attributes:
 - i) Nominal
 - ii) Ordinal
 - iii) Interval
 - iv) Ratio
- Nominal and ordinal attributes are collectively referred to as **categorical or qualitative** attributes.
- Interval and ratio types attributes are collectively referred to as **numeric or quantitative** attributes

Categorical or Qualitative attributes

1. **Nominal attribute:** The values of this type of attribute are just different names, i.e., nominal values provide only enough information to distinguish one object from another.
 - Example: zip code, employee id, color, profession, gender, etc.
 - The operations we use are mode, entropy, contingency correlation, Chi square test.

Categorical or Qualitative attributes

2. Ordinal attribute: The values of an ordinal attribute provide enough information to order objects.

- Example: values from an ordered set, e.g., military or academic rank, grades, street numbers, etc.
- The operations we use are median, percentiles, rank correlation, run tests, sign tests.
- **Run test** is a statistical procedure which determines whether a sequence of data within a given distribution have been derived with a random process or not. It may be applied to test the randomness of data in a survey that collect data from an ordered population.
- The **Sign test** is a non-parametric test that is used to test whether or not two groups are equally sized. The sign test is used when dependent samples are ordered in pairs, where the bivariate random variables are mutually independent. It is based on the direction of the plus and minus sign of the observation, and not on their numerical magnitude.

Numeric or Quantitative attributes

1. **Interval attribute:** For this type of attributes, the differences between values are meaningful, i.e., a unit of measurement exists.
 - Example: calendar dates, temperature in Celcius or Fahrenheit, etc.
 - The operations we use are mean, standard deviation, Pearson's correlation, t-test, F-test.

Numeric or Quantitative attributes

- 2. Ratio attribute:** For this type of attributes, both differences and ratios are meaningful.
- Example: temperature in Kelvin, monetary quantities, counts, age, mass, length, electrical current, etc.
 - The operations we use are geometric mean, harmonic mean, percent variation.

Attribute types based on number of values

1. Discrete: A discrete attribute has a finite or countably infinite set of values.

- Example: zip code, ID number --- categorical
counts --- numeric
- Discrete attributes are often represented using integer variables
- Binary attributes: Binary attributes are a special case of discrete attributes and assume only two values, e.g., true/false, yes/no, male/female, or 0/1.
- Binary attributes are often represented as Boolean variables, or as integer variables that only take the values 0 or 1.

Attribute types based on number of values

2. Continuous: A continuous attribute is one whose values are real numbers.

- Example: temperature, height, weight, etc.
- Continuous attributes are typically represented as floating-point variables.
- Practically, real values can only be measured and represented with limited precision.
- In theory, any of the measurement scale types – nominal, ordinal, interval, and ratio – could be combined with any of the types based on the number of attribute values – binary, discrete, and continuous.

Discrete and Continuous Attributes

- Typically, nominal and ordinal attributes are binary or discrete, while interval and ratio attributes are continuous.
- However, count attributes, which are discrete, are also ratio attributes.

Asymmetric Attributes

- For asymmetric attributes, only presence of a non-zero attribute value is regarded as important.
- Consider a dataset, where each object is a student and each attribute records whether or not a student took a particular course at a university.
- For a specific student, an attribute has a value of 1 if the student took the course associated with that attribute and a value 0 otherwise.
- Because students take only a few number of all available courses, most of the values in such a dataset would be 0.
- Therefore, it is more meaningful and more efficient to focus on the non-zero values.

Asymmetric Attributes

- To illustrate, if students are compared on the basis of the courses they don't take, then most students would seem very similar, at least if the number of courses is large.
- Binary attributes where only non-zero values are important are called **asymmetric binary attributes**. This type of attribute is particularly important for association analysis.
- It is also possible to have discrete or continuous asymmetric features. For instance, if the number of credits associated with each course is recorded, then the resulting data set will consist of **asymmetric discrete or continuous attributes**.

Discretization

- Discretization:
 - Divide the range of a continuous attribute into intervals
 - Some classification algorithms only accept categorical attributes.
 - Reduce data size by discretization
 - Prepare for further analysis

Discretization and Concept Hierarchy

- Discretization
 - Reduce the number of values for a given **continuous attribute** by dividing the range of the attribute into intervals
 - Interval labels can then be used to replace actual data values
 - Supervised vs. unsupervised
 - Split (top-down) vs. merge (bottom-up)
 - Discretization can be performed recursively on an attribute
- Concept hierarchy formation
 - Recursively reduce the data by collecting and replacing low level concepts (such as numeric values for age) by higher level concepts (such as young, middle-aged, or senior)

Discretization and Concept Hierarchy Generation for Numeric Data

- Typical methods: All the methods can be applied recursively
 - Binning (covered earlier)
 - Top-down split, *unsupervised*,
 - Histogram analysis (covered earlier)
 - Top-down split, *unsupervised*
 - Clustering analysis (covered earlier)
 - Either top-down split or bottom-up merge, unsupervised
 - Entropy-based discretization: *supervised*, top-down split
 - Segmentation by natural partitioning: top-down split, *unsupervised*

Entropy-Based Discretization

- Let attribute A be a continuous-valued attribute
- Must determine the *best split point* for A
 - Sort the value A in increasing order
 - Typically, the midpoint between each pair of adjacent values is considered as a possible *split point*
 - $(a_i + a_{i+1})/2$ is the midpoint between the values of a_i and a_{i+1}
 - The point with the *minimum expected information requirement* for A is selected as the split-point for A
- Split: D_1 is the set of tuples in D satisfying $A \leq \text{split-point}$, and D_2 is the set of tuples in D satisfying $A > \text{split-point}$

Entropy-Based Discretization

- An information-based measure called entropy is used to recursively partition the values of a numeric attribute A , resulting in a hierarchical discretization.
- Given a set of data tuples, S . The entropy-based discretization of A is as follows:
 1. Each possible split point of A can be considered a potential interval boundary or threshold T . For example, a split point value v of A is considered as T that can partition the samples in S into two subsets, S_1 and S_2 satisfying the conditions $A < T$ and $A \geq T$, respectively, thereby creating a binary discretization.

Entropy-Based Discretization

2. Given S , the threshold value selected is the one that maximizes the information gain resulting from the subsequent partitioning S_1 and S_2 .
3. Let S_1 = subset of S satisfying the conditions $A < T$ and S_2 = subset of S satisfying the conditions $A \geq T$. The entropy of a set is calculated based on the class distribution of the samples in the set.
4. Let, for set S_1 , the samples are of classes, $C_1, C_2, \dots, C_m$ and n_i be the number of samples of class C_i in S_1 . Then the probability of C_i in S_1 is $p_i = \frac{n_i}{|S_1|}$.
5. The entropy of S_1 , i.e., expected information needed to classify a given sample in S_1 is

$$Ent(S_1) = - \sum_{i=1}^m p_i \log_2(p_i)$$

Entropy-Based Discretization

- The value of $Ent(S_2)$ can be computed similarly.
- The entropy of S, or expected information needed to classify a given sample in S based on the partitioning of S into subsets by the value T (i.e., split point value v of A) is given by:
$$E(S, T) = \frac{|S_1|}{|S|} Ent(S_1) + \frac{|S_2|}{|S|} Ent(S_2)$$
- The entropy of S , i.e., expected information needed to classify a given sample in S based on the partitioning of S into subsets based on class variable is given by:
$$Ent(S) = - \sum_{i=1}^m p_i \log_2(p_i) , \text{ where } p_i = \frac{n_i}{|S|} , n_i \text{ is the number of samples of class } C_i \text{ in } S.$$
- The encoding information gained partitioning by value T (i.e., value v of A) is $Gain(A, T) = Ent(S) - E(S, T)$

Entropy-Based Discretization

- The value v of A for which information gain is maximum is considered as the threshold T and the attribute A is discretized into two intervals, $A < T$ and $A \geq T$
- S_1 is the set for which $A < T$ and S_2 is the set for which $A \geq T$.
- The same process is repeated in S_1 and S_2 for determining the threshold values.
- The process of discretization is continued until some stopping criterion is met, such as:
 - i) predefined number of intervals are obtained
 - ii) $\text{Ent}(S) - E(S, T) > \delta$ for the set S

Entropy-Based Discretization

- Entropy-based discretization uses class information. This makes it more likely that the interval boundaries are defined to occur in places that may help improve classification accuracy.
- Exercise: Discretize Age based on the following:

Sample	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Age	30	30	35	45	45	45	35	30	30	45	30	35	35	45
Class	no	no	yes	yes	yes	no	yes	no	yes	yes	yes	yes	yes	no

Segmentation by Natural Partitioning

- **We need to know about Percentile**
- **Percentile:** The percentile of data value from a set of data values is a statistical measure that gives the percentage of data values that fall below a given data value.
- The percentile formula determines the performance of a person over others. The percentile is used in finding where a student stands on a test compared to other candidates.
- For example, in a group of 20 students, X scores 4th highest and 80% of the students score less than him. It means that X is at the 80th percentile.
- Often percentile and percentage are taken as one but both are different concepts.

Percentage and Percentile

- A percentage is where the **fraction** is considered as one term while a percentile of a value is the **percentage of the values** that are below the given values out of the whole set of values.
- Example: If an exam is conducted out of 100 marks, then:
 - (i) we say that a student scored 100 "percent" if and only if he had scored $100/100$.
 - (ii) we say that a student scored 100 "percentile" if all the students (100% students) scored less than him.

Percentile Formula

- Percentile formulas are usually helpful in grading test scores or biometric measurements. Hence, the percentile formula is:

$P = (n/N) \times 100$ where, n = ordinal rank of the given value or value below the number, N = number of values in the data set, and P = percentile.

- The percentile of x is defined as

Percentile = (Number of Values Below “x” / Total Number of Values) $\times$ 100

Percentile Calculation

- **Step 1:** Collect the data set
- **Step 2:** Arrange the data set in ascending order
- **Step 3:** Determine the total number of observations
- **Step 4:** Identify the data value for which you are interested to find the percentile
- **Step 5:** Count the number of data values that are less than the above value
- **Step 6:** Divide the number from **Step 5** by the number from **Step 3** to find the percentile of the given data value

Examples Using Percentile Formula

- **Ex-1:** The scores obtained by 10 students are 38, 47, 49, 58, 60, 65, 70, 79, 80, 92. Using the percentile formula, calculate the percentile for score 70?
- **Solution:**
 - (i) Scores obtained by students are 38, 47, 49, 58, 60, 65, 70, 79, 80, 92
 - (ii) Number of scores below 70 = 6
 - (iii) Percentile of 70 = $(\text{Number of Values Below 70} / \text{Total Number of Values}) \times 100 = (6/10) \times 100 = 0.6 \times 100 = 60$
 - (iv) Therefore, the percentile for score 70 = 60%

Examples Using Percentile Formula

- **Ex-2:** The weights of 10 people were recorded in kg as 35, 41, 42, 56, 58, 62, 70, 71, 90, 77. How to find percentile for the weight 58 kg?

- **Solution:**

(i) Weight of the people are 35, 41, 42, 56, 58, 62, 70, 71, 77, 90

(ii) Number of people with weight below 58 kg = 4

(iii) Percentile for weight 58 kg = (Number of Values Below 58 kg / Total Number of Values) $\times$ 100 = (4/10) $\times$ 100 = 0.4 $\times$ 100 = 40%

(iv) Therefore, the percentile for weight 58 kg = 40%

Examples Using Percentile Formula

- **Ex-3:** The scores of 10 students are 56, 45, 69, 78, 72, 94, 82, 80, 63, 59. Using the percentile formula, find the 70th percentile.
- **Solution:** (i) Let x is the 70th percentile
(ii) Arrange the data in ascending order : 45, 56, 59, 63, 69, 72, 78, 80, 82, 94.
(iii) Percentile = (Number of Values Below "x" / Total Number of Values) $\times$ 100 $\Rightarrow$ Number of Values Below "x" = 7
(iv) Counting 7 values from left to right we reach 80, and we can say that all the values below 80 will come under the 70th percentile.
(v) In other words, 70% of the values are below 80. Therefore, the 70th percentile is 80.

Segmentation by Natural Partitioning

- A simply 3-4-5 rule can be used to segment numeric data into relatively uniform, “natural” intervals. For ex., annual salaries broken into ranges like (\$50,000, \$60,000] are often more desirable than ranges like (\$51,263.98, \$60,872.34], obtained by some sophisticated clustering analysis.
- In general, the rule partitions a given range of data into 3, 4, or 5 relatively equiwidth intervals, recursively and level by level, based on the value range at the most significant digit.

Segmentation by Natural Partitioning

- The rule is as follows:
 - If an interval covers 3, 6, 7 or 9 distinct values at the most significant digit, partition the range into 3 intervals (3 equiwidth intervals for 3, 6, 9, and 3 intervals in the grouping of 2-3-2 for 7).
 - If it covers 2, 4, or 8 distinct values at the most significant digit, partition the range into 4 equiwidth intervals
 - If it covers 1, 5, or 10 distinct values at the most significant digit, partition the range into 5 equiwidth intervals

Segmentation by Natural Partitioning

- The rule can be recursively applied to each interval, creating a concept hierarchy for the given numeric attribute.
- Since there could be some dramatically large positive or negative values in a data set, the top-level segmentation, based merely on the minimum and maximum values, may derive distorted result.
- For ex., the assets of a few people could be several orders of magnitude higher than those of others in the data set.

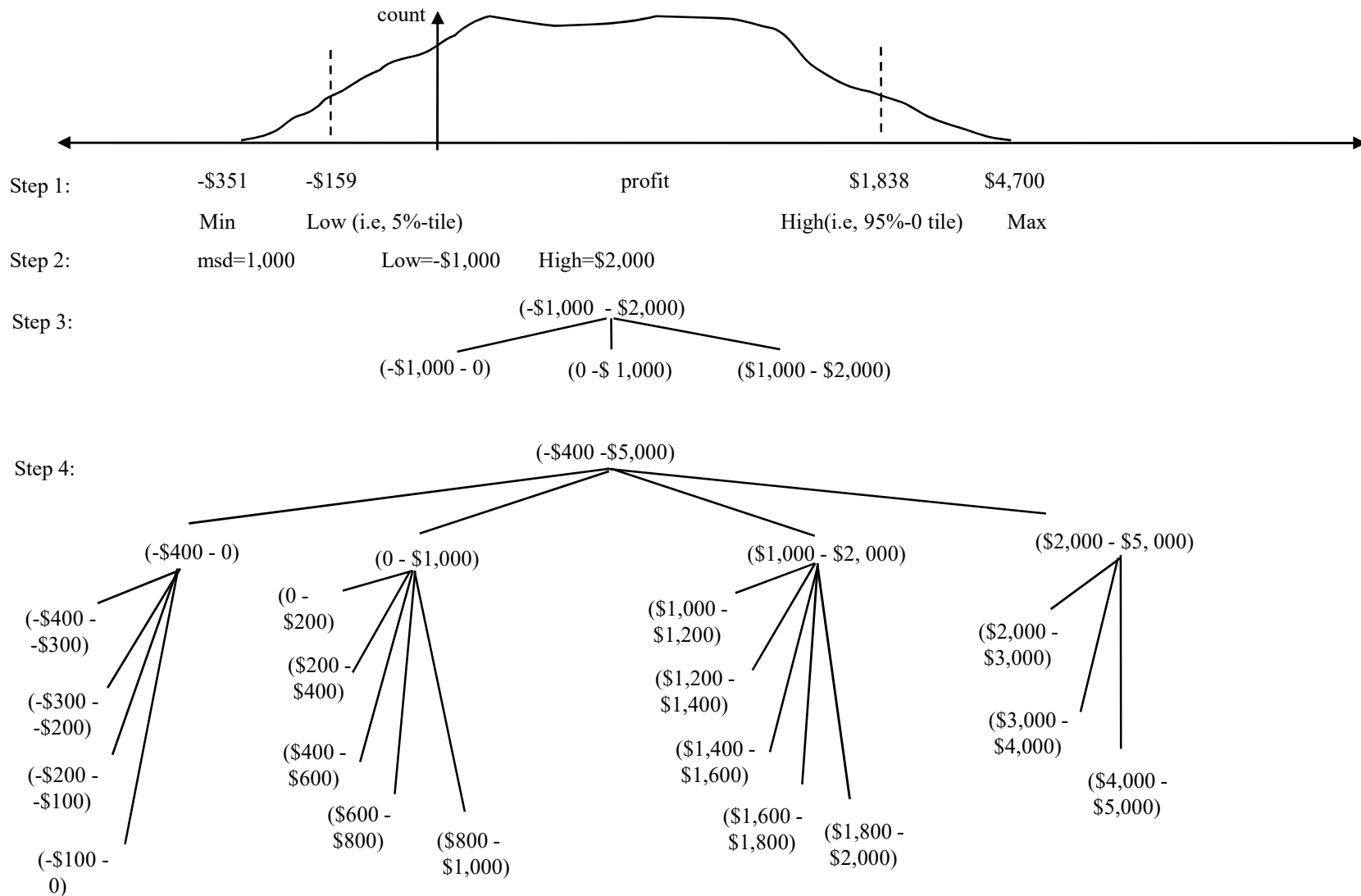
Segmentation by Natural Partitioning

- So segmentation based on the maximum asset values may lead to a **highly biased hierarchy**.
- Thus the top level segmentation can be performed based on the range of data values representing the majority (e.g., 5th percentile to 95th percentile) of the given data.
- The extremely high or low values beyond the top-level segmentation will form distinct interval(s) that can be handled separately, but in a similar manner.

Example of a Segmentation by Natural Partitioning (3-4-5 rule)

- Suppose the profit of different branches of an AllElectronics for the year 2020 cover a wide range, from -\$351.00 to \$4700.
- A user wishes to have a concept hierarchy for profit automatically generated. Let, we use the interval as (l, r) .
- Suppose the data within the 5th percentile and 95th percentile are -\$159 and \$1838. The results of applying the 3-4-5 rule are shown in the following slide.

Example of 3-4-5 Rule



Example of a Segmentation by Natural Partitioning (3-4-5 rule)

- Based on the given information, the minimum and maximum values are, $\text{Min} = -\$351$, and $\text{Max} = \$4700$.
- The low (5th percentile) and high (95th percentile) values to be considered for the top or first level of segmentation are $\text{Low} = -\$159$, and $\text{High} = \$1838$.
- Given Low and High, the most significant digit is at the thousand digit position (i.e., $\text{msd} = 1000$).
- Rounding Low down to the thousand digit, we get $\text{Low}' = -\$1000$; rounding High up to the thousand digit, we get $\text{High}' = \$2000$;

Example of a Segmentation by Natural Partitioning (3-4-5 rule)

- This interval ranges over three distinct values at the most significant digit, that is, $(\text{High}' - \text{Low}')/\text{msd} = (2000 - (-1000))/1000 = 3$.
- So the segment is partitioned into three equiwidth subsegments according to the 3-4-5 rule: $(-\$1000, \$0)$, $(\$0, \$1000)$, and $(\$1000, \$2000)$. This represent the **top tier of the hierarchy**.

Example of a Segmentation by Natural Partitioning (3-4-5 rule)

- We now examine the Min and Max values to see how they “fit” into the first level partitions.
- Since the first interval ($-\$1000, \0) covers the $\text{Min} = \$-351$, i.e., $\text{Low}' < \text{Min}$, we can adjust the left boundary of this interval to make the interval smaller.
- The most significant digit of Min is the hundred digit position. Rounding Min down to this position, we get $\text{Min}' = -\$400$. Therefore, the first interval is redefined as $(-\$400, \$0)$

Example of a Segmentation by Natural Partitioning (3-4-5 rule)

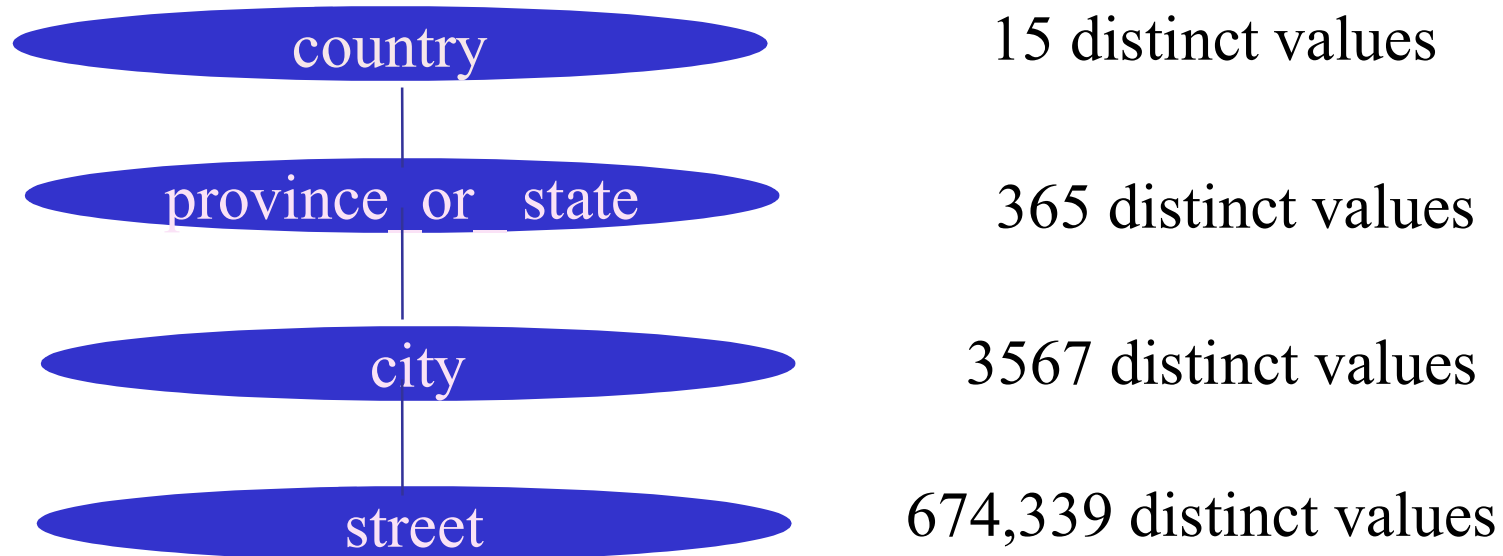
- Since the last interval, (\$1000, \$2000), does not cover the Max value (Max=4700), i.e., $\text{Max} > \text{High}'$, we need to create a new interval to cover it.
- Rounding up Max at its most significant digit position, the new interval is (\$2000, \$5000).
- Hence, the topmost level of the hierarchy contains four partitions, (-\$400, \$0), (\$0, \$1000), (\$1000, \$2000), and (\$2000, \$5000).
- Recursively, each interval can be further partitioned according to the 3-4-5 rule to form the next lower level of the hierarchy.
- The 3-4-5 rule can be carried on iteratively at deeper levels as necessary.

Concept Hierarchy Generation for Categorical Data

- Specification of a partial/total ordering of attributes explicitly at the schema level by users or experts
 - street < city < state < country
- Specification of a hierarchy for a set of values by explicit data grouping
 - {Urbana, Champaign, Chicago} < Illinois
- Specification of only a partial set of attributes
 - E.g., only street < city, not others
- Automatic generation of hierarchies (or attribute levels) by the analysis of the number of distinct values
 - E.g., for a set of attributes: {street, city, state, country}

Automatic Concept Hierarchy Generation

- Some hierarchies can be automatically generated based on the analysis of the number of distinct values per attribute in the data set
 - The attribute with the most distinct values is placed at the lowest level of the hierarchy
 - Exceptions, e.g., weekday, month, quarter, year



Data Preprocessing

- Why preprocess the data?
- Data cleaning
- Data integration and transformation
- Data reduction
- Discretization and concept hierarchy generation
- Summary

Summary

- Data preparation or preprocessing is a big issue for both data warehousing and data mining
- Descriptive data summarization is needed for quality data preprocessing
- Data preparation includes
 - Data cleaning and data integration
 - Data reduction and feature selection
 - Discretization
- A lot of methods have been developed but data preprocessing still an active area of research

References

- D. P. Ballou and G. K. Tayi. Enhancing data quality in data warehouse environments. *Communications of ACM*, 42:73-78, 1999
- T. Dasu and T. Johnson. *Exploratory Data Mining and Data Cleaning*. John Wiley & Sons, 2003
- T. Dasu, T. Johnson, S. Muthukrishnan, V. Shkapenyuk. [Mining Database Structure; Or, How to Build a Data Quality Browser](#). *SIGMOD' 02*.
- H.V. Jagadish et al., Special Issue on Data Reduction Techniques. *Bulletin of the Technical Committee on Data Engineering*, 20(4), December 1997
- D. Pyle. *Data Preparation for Data Mining*. Morgan Kaufmann, 1999
- E. Rahm and H. H. Do. Data Cleaning: Problems and Current Approaches. *IEEE Bulletin of the Technical Committee on Data Engineering*. Vol.23, No.4
- V. Raman and J. Hellerstein. *Potters Wheel: An Interactive Framework for Data Cleaning and Transformation*, *VLDB' 2001*
- T. Redman. *Data Quality: Management and Technology*. Bantam Books, 1992
- Y. Wand and R. Wang. Anchoring data quality dimensions ontological foundations. *Communications of ACM*, 39:86-95, 1996
- R. Wang, V. Storey, and C. Firth. A framework for analysis of data quality research. *IEEE Trans. Knowledge and Data Engineering*, 7:623-640, 1995

Thank You!